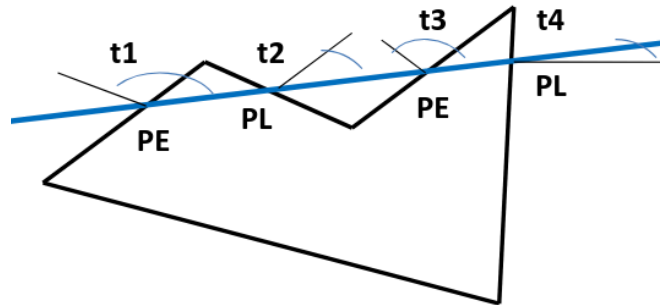


SET A

1. Window: $X_{min} = -40$, $X_{max} = -40 + 80 = 40$
 $Y_{min} = 20 - 50 = -30$, $Y_{max} = 20$
 - a. $(0, 0)$, outcode = 0000
 - b. $(50, 20)$, outcode = 0010
 - c. $(20, 50)$, outcode = 1000
 - d. $(-50, -30)$, outcode = 0001
 - e. $(50, -40)$, outcode = 0110

2. The algorithm fails in the case of the Concave window. If we observe the polygon drawn to the right, we get $t_L = t_2$ and $t_E = t_3$. As $t_E > t_L$, the algorithm will clip the entire line which is wrong.



3. $P1 = (-2, 3)$, $P2 = (4, -5)$
 $P(t) = P1 + t(P2 - P1) = (-2, 3) + t(4+2, -5-3) = (-2+6t, 3-8t)$
The mid point is $P(0.5) = (-2+6*0.5, 3-8*0.5) = (1, -1)$

4. Window: $X_{min} = 0$, $X_{max} = 40$
 $Y_{min} = -80$, $Y_{max} = 0$
Equation of line, $(y1-y) / (y1 - y2) = (x1-x) / (x1-x2)$
 $\Rightarrow (5-y) / (5 + 110) = (-10-x) / (-10-50)$
 $\Rightarrow -12 * (5-y) = (-10-x) * 23$
 $\Rightarrow y = (-23/12)x - 85/6$
 $\Rightarrow x = (-12/23)y - 170/23$

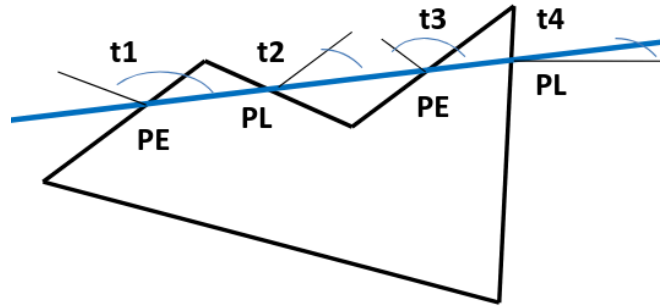
P1	P2	oc1	oc2	oc1 & oc2	Decision	Intersecting Point
$(-10, 5)$	$(50, -110)$	1001	0110	0000	Partially Accepted	$Y=0$ $X=(-12/23)*0 - 170/23$ $=-170/23=-7.4$
$(-7.4, 0)$	$(50, -110)$	0001	0110	0000	Partially Accepted	$X=0$ $Y=(-23/12)*(0) - 85/6$ $=-14.2$
$(0, -14.2)$	$(50, -110)$	0000	0110	0000	Partially Accepted	$Y=-80$ $X=(-12/23)*-80 - 170/23$ $=790/23=34.35$
$(0, -14.2)$	$(34.35, -80)$	0000	0000	0000	Accepted	

SET B

1. Window: $X_{min} = -20$, $X_{max} = -20 + 80 = 60$
 $Y_{min} = 40 - 50 = -10$, $Y_{max} = 40$

- f. $(0, 0)$, outcode = 0000
- g. $(-20, 50)$, outcode = 1000
- h. $(50, -20)$, outcode = 0100
- i. $(-30, 50)$, outcode = 1001
- j. $(40, -50)$, outcode = 0100

2. The algorithm fails in the case of the Concave window. If we observe the polygon drawn to the right, we get $t_L = t_2$ and $t_E = t_3$. As $t_E > t_L$, the algorithm will clip the entire line which is wrong.



3. $P1 = (-3, 2)$, $P2 = (5, -4)$

$$P(t) = P1 + t(P2 - P1) = (-3, 2) + t(5+3, -4-2) = (-3+8t, 2-6t)$$

$$\text{The mid point is } P(0.5) = (-3+8*0.5, 2-6*0.5) = (1, -1)$$

4. Window: $X_{min} = 0$, $X_{max} = 80$
 $Y_{min} = -40$, $Y_{max} = 0$

$$\text{Equation of line, } (y1-y) / (y1 - y2) = (x1-x) / (x1-x2)$$

$$\Rightarrow (10-y) / (10 + 50) = (-5-x) / (-5-110)$$

$$\Rightarrow -23 * (10-y) = (-5-x) * 12$$

$$\Rightarrow y = (-12/23)x + 170/23$$

$$\Rightarrow x = (-23/12)y + 85/6$$

P1	P2	oc1	oc2	oc1 & oc2	Decision	Intersecting Point
$(-5, 10)$	$(110, -50)$	1001	0110	0000	Partially Accepted	$Y=0$ $X=(-23/12)*0 + 85/6$ $=85/6=14.17$
$(14.17, 0)$	$(110, -50)$	0000	0110	0000	Partially Accepted	$Y=-40$ $X=(-23/12)*-40+85/6$ $=545/6=90.83$
$(14.17, 0)$	$(90.83, -40)$	0000	0010	0000	Partially Accepted	$X=80$ $Y=(-12/23)*80 + 170/23$ $=-790/23=-34.35$
$(14.17, 0)$	$(80, -34.35)$	0000	0000	0000	Accepted	